



IoT-Based Weed Detection & Removal Robot

Phase 4 — Master Project Plan

Full System Design · Task Distribution · Testing Checklist · Completion Criteria

Infinity Arc Studio | Version 1.0

1. Project Overview

Phase 4 — Build a fully controllable 4WD Weed-Removal Robot

This master plan consolidates all Phase 4 objectives, task assignments, hardware requirements, system architecture, development workflow, testing criteria, and completion standards into a single reference document. Phase 4 focuses on building a fully controllable 4WD robot — **no AI or camera integration** is included in this phase (reserved for Phase 5).

Feature	Status
Forward / Backward / Left / Right movement	Required
Stop command	Required
Cutter Motor ON / OFF	Required
Obstacle Detection (HC-SR04)	Required
Dashboard Control UI	Required
Backend REST APIs	Required
ESP32 Firmware & Wi-Fi	Required
Activity Logging	Required
AI / Camera Integration	Phase 5 only

2. System Architecture

Component	Role
Dashboard (Browser UI)	User sends movement & cutter commands via the web dashboard.
Backend APIs (Node.js)	REST endpoints receive commands and forward them to the ESP32.
ESP32 DevKit V1	Microcontroller receives commands over Wi-Fi and drives hardware.
L298N Motor Drivers (×2)	Dual H-bridge ICs control the 4 DC gear motors independently.
DC Gear Motors (×4)	Drive wheels — 2 left, 2 right — for directional movement.
Relay Module	Switches the RS775 cutter motor ON or OFF on command.
HC-SR04 Sensors (×2)	Ultrasonic sensors detect obstacles and trigger emergency stop.

Dashboard → Backend APIs → ESP32 → L298N Drivers → 4 Wheel Motors

Dashboard → Backend APIs → ESP32 → Relay Module → RS775 Cutter Motor

3. Hardware Components Required

#	Component	Purpose
1	ESP32 DevKit V1	Main microcontroller — Wi-Fi + GPIO
2	L298N Motor Driver	×2 — Controls 4 DC motors in pairs
3	4WD Chassis Kit	Chassis frame with wheel mounts
4	DC Gear Motors	×4 — One per wheel
5	RS775 Cutter Motor	High-torque motor for weed cutting
6	Relay Module	Switches cutter motor power
7	HC-SR04 Ultrasonic Sensor	×2 — Front obstacle detection
8	12V Battery Pack	Main power source
9	LM2596 Buck Converter	Steps down voltage for logic circuits
10	Emergency Stop Switch	Hardware safety cutoff
11	Jumper Wires & Connectors	Wiring kit for all connections
12	Mounting Hardware	Screws, standoffs, brackets

4. Team Task Distribution

The project is divided among four members. Each member owns specific files and deliverables. All members share responsibility for hardware assembly.

Member 1

Backend Developer

Primary Folder:

[software/backend/](#)

Files:

- server.js
- routes/*
- data/logs.json
- package.json

Tasks:

1. Design robot command API structure
2. Implement movement APIs: /forward, /backward, /left, /right, /stop
3. Implement cutter APIs: /cutter/on, /cutter/off
4. Create robot status endpoint
5. Build activity logging APIs
6. Define request/response JSON formats
7. Support integration testing with ESP32

Deliverable:

[Working backend APIs + API documentation](#)

Member 2

Frontend Developer

Primary Folder:

[software/frontend/](#)

Files:

- index.html
- pages/logs.html
- pages/settings.html
- js/api.js
- js/script.js
- js/logs.js
- js/settings.js
- css/style.css
- css/logs.css
- css/settings.css

Tasks:

1. Build robot control dashboard UI
2. Add directional movement buttons (F/B/L/R/Stop)
3. Add cutter ON/OFF controls
4. Display live robot status
5. Display activity logs page
6. Handle API errors and connection states
7. Implement mobile-responsive design

Deliverable:

[Fully functional control dashboard](#)

Member 3 ESP32 Developer	
Primary Folder: hardware/ Files: - ESP-32_sketch_apr12a/sketch_apr12a.ino - components-list.md	Tasks: 1. Configure Wi-Fi communication on ESP32 2. Integrate two L298N motor drivers 3. Implement forward / backward / left / right / stop motor logic 4. Integrate relay control for RS775 cutter motor 5. Integrate HC-SR04 ultrasonic sensors 6. Implement obstacle detection & auto-stop logic 7. Receive and parse commands from backend 8. Send robot status updates back to backend Deliverable: Complete ESP32 firmware

Member 4 QA, Integration & Documentation	
Primary Folder: docs/ Files: - wiring-diagrams/ - architecture-diagrams/ - testing-reports/ - bug-tracker/	Tasks: 1. Maintain task tracking board 2. Prepare wiring and architecture diagrams 3. Verify all hardware connections 4. Create and maintain testing checklist 5. Perform API integration testing 6. Perform movement and cutter testing 7. Track bugs and fixes 8. Prepare final documentation and demo Deliverable: Test reports + complete project documentation

5. Shared Responsibility — Hardware Assembly

All four members participate equally

Step 1	Chassis assembly — mount frame and structural components
Step 2	Motor installation — attach and secure all 4 DC gear motors
Step 3	Battery mounting — install 12V battery pack securely
Step 4	ESP32 installation — mount and connect microcontroller
Step 5	L298N installation — mount both motor driver boards
Step 6	Relay installation — wire relay module for cutter control
Step 7	Cutter assembly — mount and connect RS775 cutter motor
Step 8	Wiring and cable management — route and secure all wires

Step 9	Final hardware verification — check all connections before power-on
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6. Robot Movement Logic

Command	Left Motors	Right Motors	Result
Forward	FORWARD	FORWARD	Robot moves straight ahead
Backward	BACKWARD	BACKWARD	Robot moves straight back
Left Turn	BACKWARD	FORWARD	Robot pivots left
Right Turn	FORWARD	BACKWARD	Robot pivots right
Stop	STOP	STOP	All motors halt
Obstacle Detected	STOP	STOP	Auto-stop when threshold reached

7. Development Workflow

Step 1	Backend APIs Completed Member 1 finishes all REST endpoints. Tested locally using Postman.
Step 2	Frontend Integrates APIs Member 2 connects dashboard buttons to the backend endpoints.
Step 3	ESP32 Firmware Consumes APIs Member 3 programs ESP32 to receive commands and drive motors/relay.
Step 4	Integration Testing Member 4 verifies the full chain: Dashboard → Backend → ESP32 → Hardware.
Step 5	Hardware Testing Physical robot tested for movement, cutter, and obstacle detection.
Step 6	Bug Fixing All reported bugs tracked and resolved by responsible member.
Step 7	Final Demonstration Complete system demo with all features verified.

8. Software Installation Requirements

Scope	Tools / Packages
All Members	Git, VS Code, Node.js
Backend (M1)	Postman, npm packages (express, cors, etc.)
Frontend (M2)	Browser, npm packages, React (optional)
ESP32 (M3)	Arduino IDE, ESP32 Board Package
Future — Phase 5	Python, OpenCV, YOLOv8, PyTorch

9. Phase 4 Testing Checklist

Movement Tests	
Robot moves Forward correctly	■ Pass ■ Fail ■ N/A
Robot moves Backward correctly	■ Pass ■ Fail ■ N/A
Robot turns Left correctly	■ Pass ■ Fail ■ N/A
Robot turns Right correctly	■ Pass ■ Fail ■ N/A
Robot stops on Stop command	■ Pass ■ Fail ■ N/A
Cutter Tests	
Cutter motor turns ON via dashboard	■ Pass ■ Fail ■ N/A
Cutter motor turns OFF via dashboard	■ Pass ■ Fail ■ N/A
Safety Tests	
Emergency stop switch halts all motors	■ Pass ■ Fail ■ N/A
Obstacle detection stops robot at threshold	■ Pass ■ Fail ■ N/A
Integration Tests	
Dashboard → Backend → ESP32 command chain works	■ Pass ■ Fail ■ N/A
ESP32 → L298N → Motors execute correctly	■ Pass ■ Fail ■ N/A
ESP32 → Relay → Cutter motor works	■ Pass ■ Fail ■ N/A
Logs generated for all commands	■ Pass ■ Fail ■ N/A
Status updates returned to dashboard	■ Pass ■ Fail ■ N/A

10. Phase 4 Completion Criteria

Phase 4 is considered complete only when **all** of the following criteria are verified:

✓ Robot moves in all four directions (Forward, Backward, Left, Right) reliably.
✓ Robot stops correctly on command and on obstacle detection.
✓ Cutter motor switches ON and OFF via the dashboard.
✓ Obstacle detection halts the robot before collision.
✓ Dashboard successfully controls all robot functions.
✓ Activity logs are generated and stored for all commands.

✓ All integration tests pass (Dashboard → Backend → ESP32 → Hardware).

✓ Hardware is securely mounted with no loose connections.

✓ Robot operates continuously for a minimum of 15 minutes without failure.

Phase 5 Roadmap: After Phase 4 completion, the project will add Python, OpenCV, YOLOv8, and PyTorch for AI-based weed detection via camera integration.